Introduction

CalicoST is a computational framework developed to infer clonal architecture and copy number variation (CNV) directly from spatial transcriptomics data. Tracing somatic evolution across tumor space and time remains a major challenge in cancer research. Spatially resolved transcriptomics measures gene expression across thousands of tumor locations but does not directly detect genomic aberrations. CalicoST infers allele-specific copy number aberrations (CNAs) and reconstruct spatial tumor evolution from spatial transcriptomic data.

Reference:
Ma C, Balaban M, Liu J, Chen S, Wilson MJ, Sun CH, Ding L, Raphael BJ. Inferring allele-specific copy number aberrations and tumor phylogeography from spatially resolved transcriptomics. Nat Methods. 2024;21(12):2239-47. Epub 20241030. doi: 10.1038/s41592-024-02438-9. PubMed PMID: 39478176; PMCID: PMC11621028.

We are running the **expression-only (exponly option) version** of CalicoST. This version does **not require allele-specific information** (e.g., SNPs or B-allele frequencies) and instead infers CNVs using:

- Gene expression counts
- Genomic gene ordering
- Spatial coordinates of spots or cells

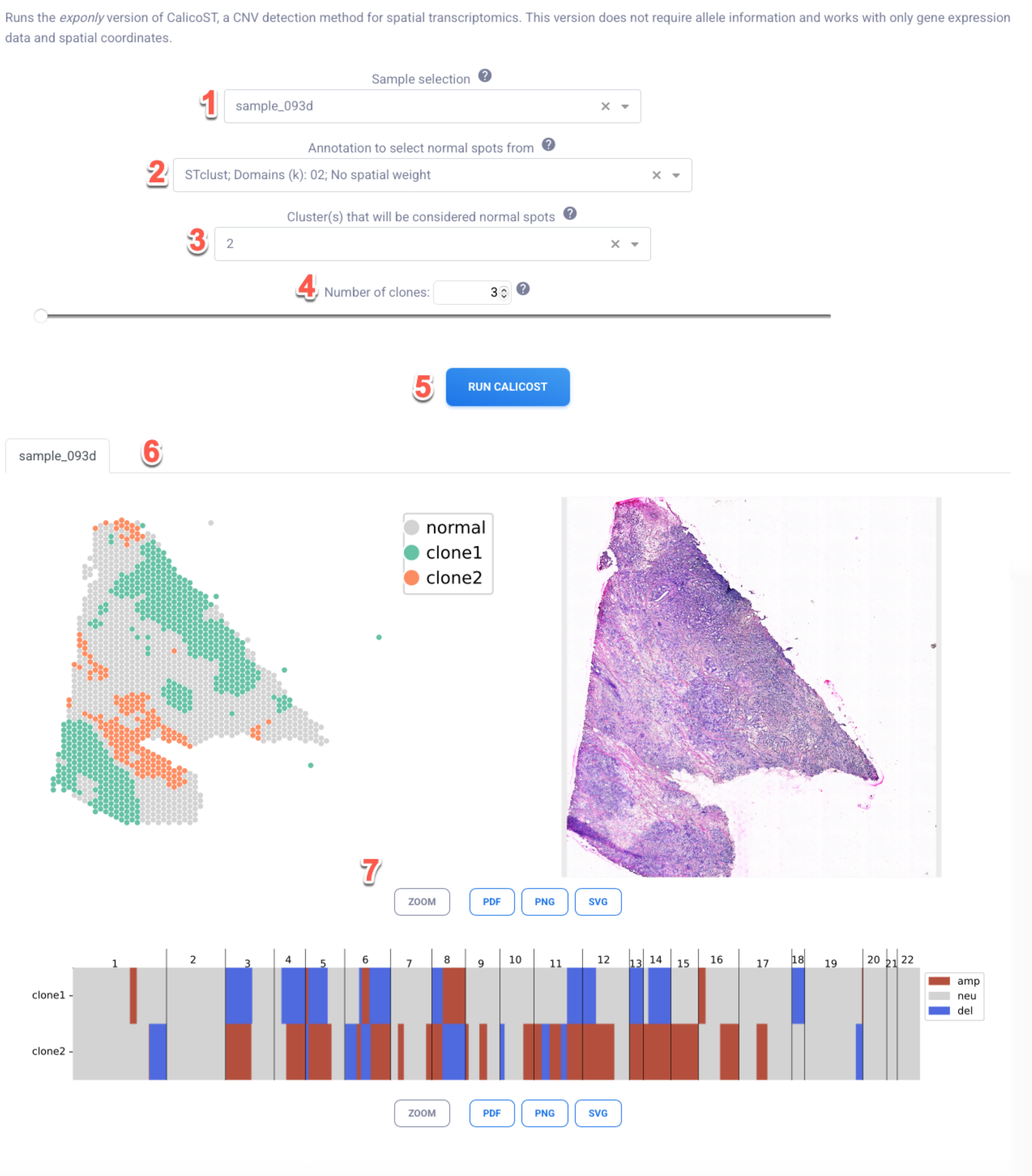
This makes it particularly suitable for platforms where allele information is such as many targeted spatial panels.

CalicoST (Expression-Only Mode) — GUI Documentation

The screenshot below illustrates the graphical user interface (GUI) for running the expression-only (exponly) version of CalicoST. This version detects copy number variation (CNV) from spatial transcriptomics data using only gene expression counts and spatial coordinates, without requiring allele-specific information.

Steps
The Steps is divided into three logical sections:

- **Input configuration (1–4)**
- **Execution (5)**
- **Outputs and visualization (6–7)**



1. Sample Selection

Select the spatial transcriptomics sample to analyze.

- Dropdown menu lists available processed samples.
- Example shown: sample_093d
- Ensure the sample has gene expression data and spatial coordinates loaded.

This defines the dataset on which CNV inference will be performed.

2. Annotation to Select Normal Spots
Choose an existing clustering or annotation layer to define candidate "normal" regions.
- Example: STclust; Domains (k): 02; No spatial weight
- This annotation is used to distinguish tumor vs. non-tumor spots.

CalicoST requires a reference population of "normal" spots to establish a baseline expression profile for CNV inference.

3. Cluster(s) Considered Normal
Specify which cluster ID(s) from the selected annotation should be treated as normal.
- Example shown: cluster 2
- Multiple clusters can be selected if needed.

These spots are assumed to represent diploid baseline expression and are used to normalize tumor CNV signals.
Proper normal selection is critical for accurate CNV calling.

4. Number of Clones
Set the expected number of tumor clones to infer.
- Example shown: 3
- This determines how many distinct CNV-based clonal groups CalicoST will attempt to identify.

Guidance:
- Use prior biological knowledge if available.
- Start with 2–3 clones for exploratory analysis.
- Increasing clone number increases model complexity.

5. Run CalicoST
Click **"RUN CALICOST"** to start the analysis.

Computation time depends on:
- Number of spots
- Number of genes
- Number of inferred clones

Output & Visualization
After execution, results appear in the lower panel.

6. Spatial Clone Map
Left Panel:

- Spatial spot map colored by inferred class:
    - **Gray** → Normal
    - **Green** → Clone 1
    - **Orange** → Clone 2

This map visualizes spatial clonal architecture across tissue.
Right Panel:

- Histology image (e.g., H&E)
- Enables visual comparison between morphology and inferred clones.

This helps assess:

- Tumor boundaries
- Invasive fronts
- Spatial coherence of clones

7. CNV Heatmap & Export Options
Bottom Panel:

- Genome-wide CNV profiles for each clone
- Chromosomes labeled (1–22)
- Color coding:
    - **Red (amp)** → Amplification
    - **Gray (neu)** → Neutral
    - **Blue (del)** → Deletion

Each row represents a clone's inferred copy number state across the genome.